

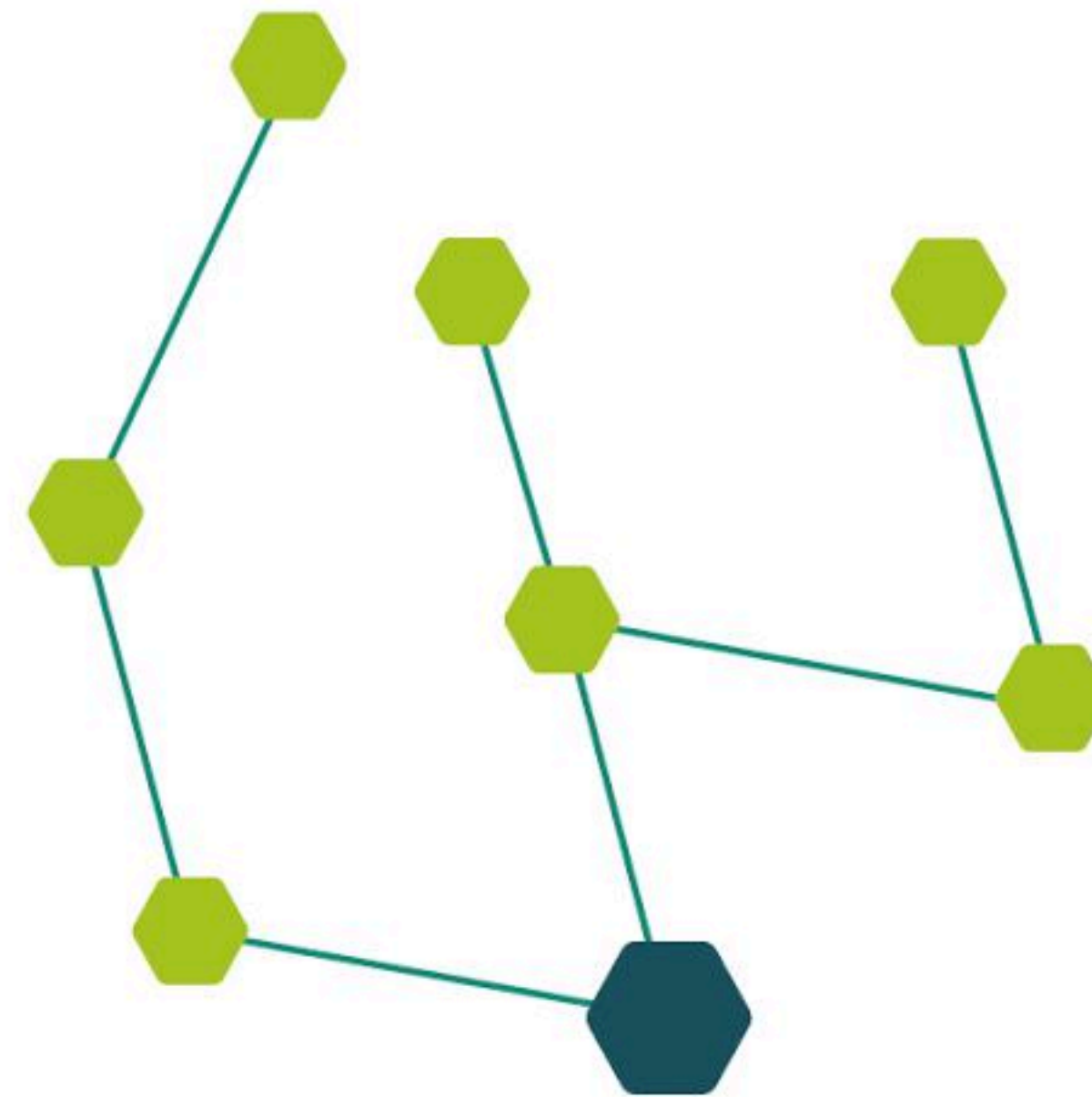
RESQ-MESH

Offline Emergency Communication & Intelligent Rescue Network

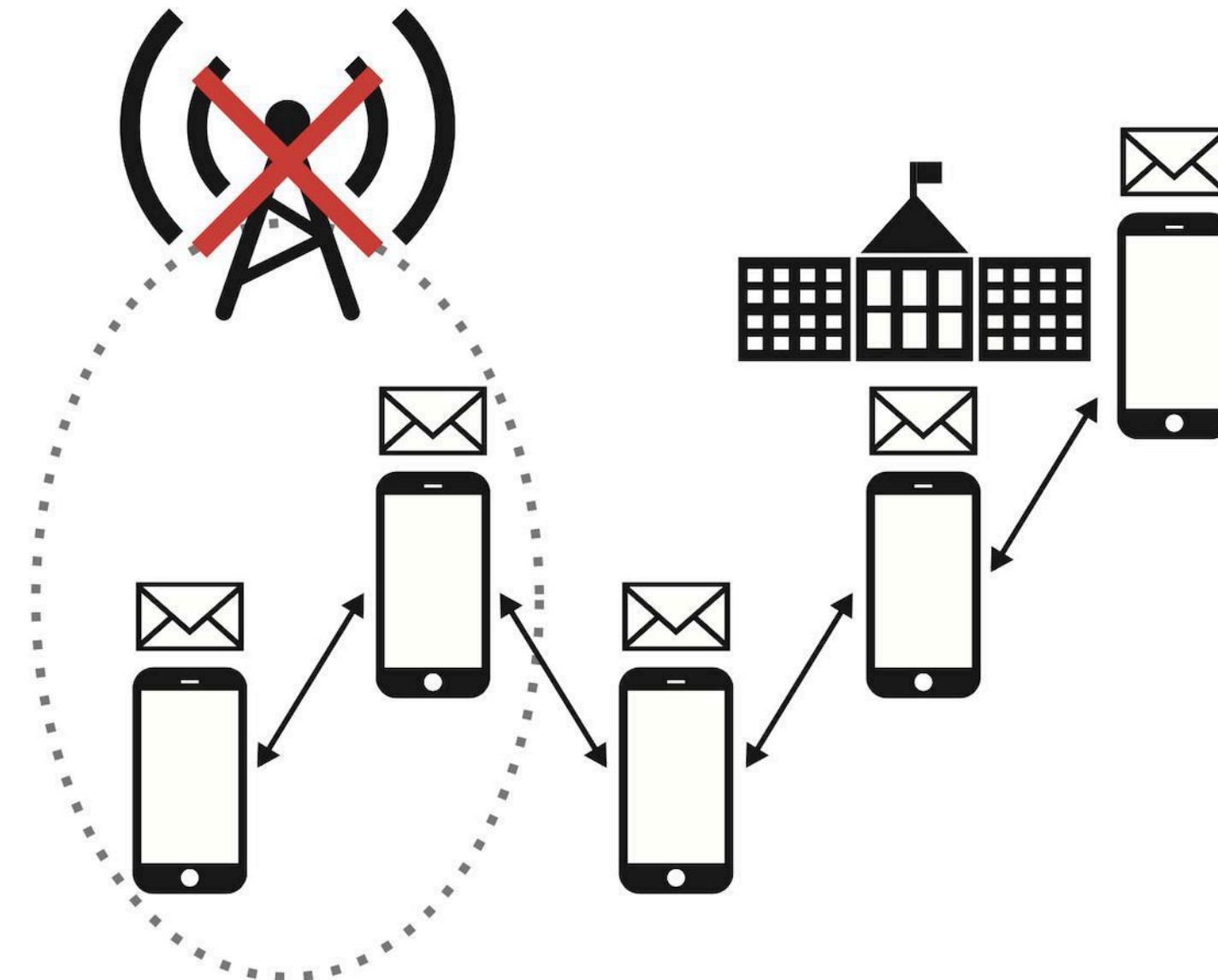
When cellular infrastructure fails, every nearby device becomes a potential lifeline.

Team Jude Force

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Mesh network



Problem Statement

During natural disasters, the **communication systems rescuers** depend on can fail at the exact moment they are needed most.

What happens next?

01 — **Connectivity Breaks** Cell towers and internet infrastructure become unavailable.

02 — **Victims Go Silent**
People trapped in affected areas cannot reliably reach emergency services.

03 — **Rescue Teams Lose Visibility**
Responders don't know where help is needed most.

04 — **Time Is Lost**
Unverified information and unsafe routes delay critical rescue operations.



Challenges & Our Approach

Building an emergency network that works when infrastructure doesn't.
Then replace the two sections on the right with:

1. Unreliable Connectivity

Challenge: Cellular networks may be unavailable during disasters.

Our approach:

Use BLE + Wi-Fi Direct + store-and-forward routing to relay SOS packets between nearby devices.

2. Limited Device Resources

Challenge: Smartphones have limited battery, processing power, and connectivity range.

Our approach:

Perform lightweight edge AI and transmit compact, encrypted SOS metadata instead of continuously sending large data.

If you can fit a third section:

3. Noisy Emergency Data

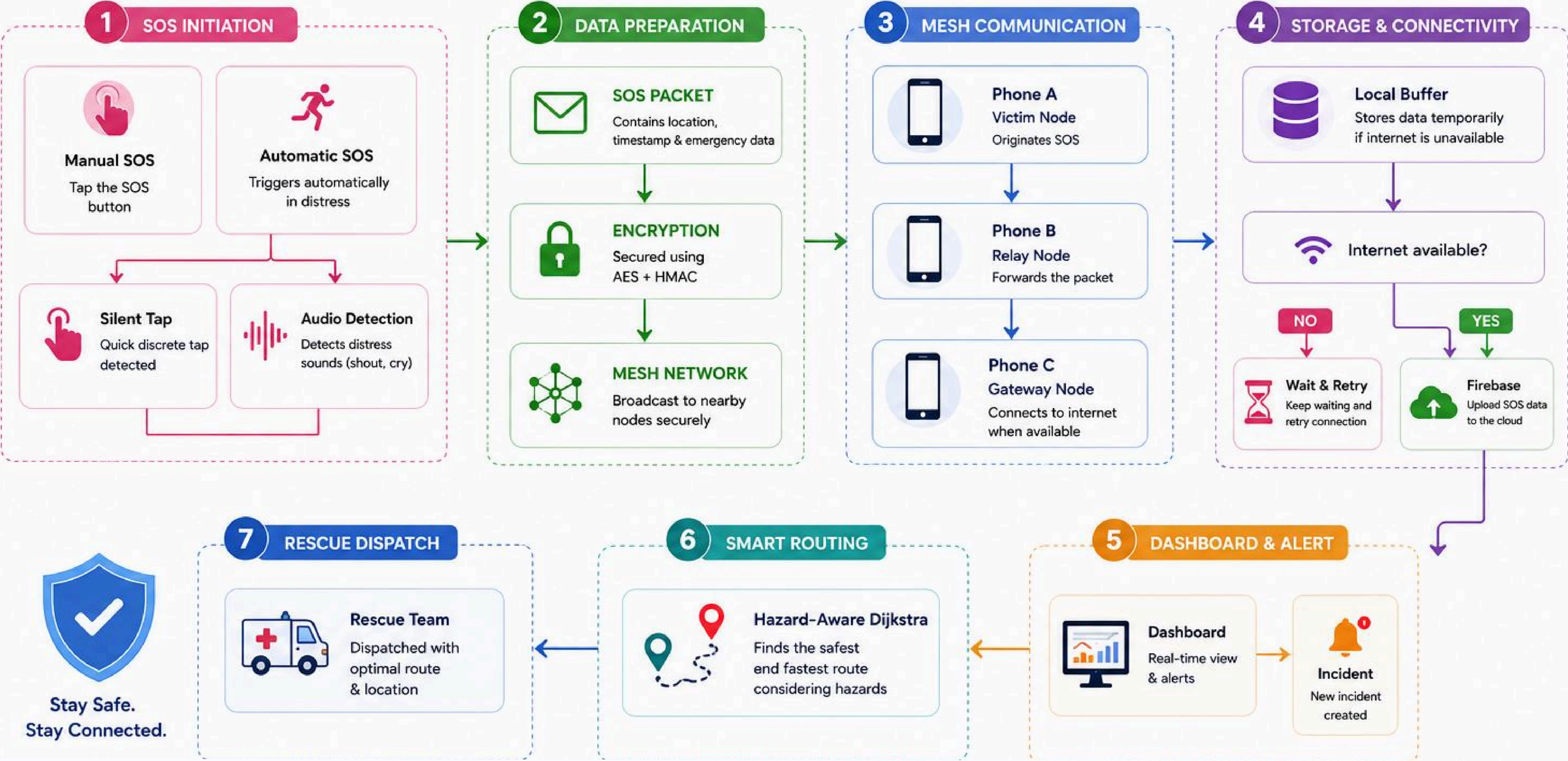
Challenge: Rescue teams may receive too many unprioritized SOS signals.

Our approach:

Use AI-assisted triage to classify distress signals and assign Critical / High / Medium / Low priorities.

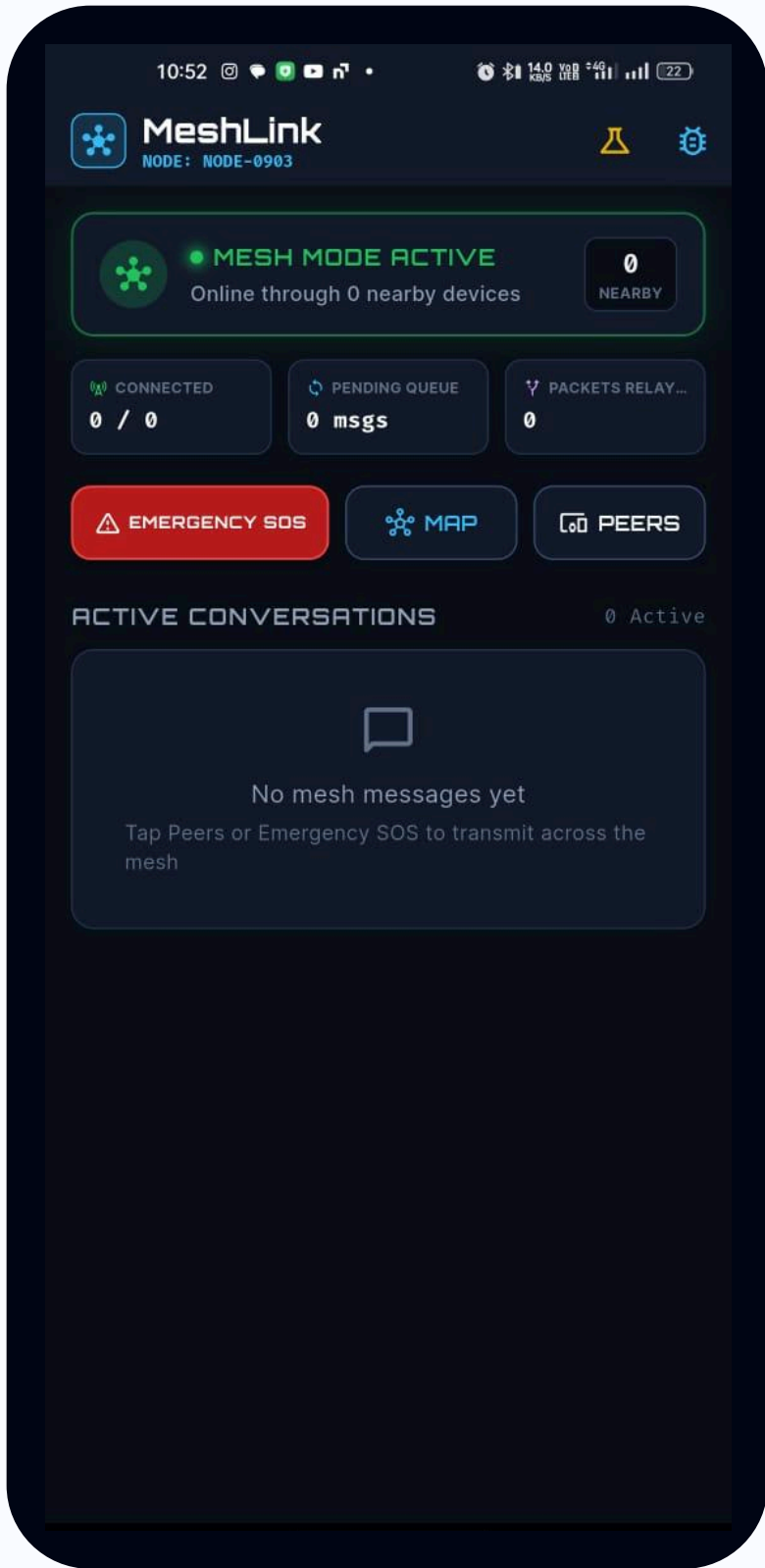


SOLUTION ARCHITECTURE



OUR SYSTEM

Home Screen



Emergency Alert Page



Nearby Phones



Home Screen

Monitor the mesh network at a glance.

Users can see active mesh status, connected peers, nearby nodes, and access emergency SOS features.

Emergency Alert Page

Send a prioritized emergency alert without cellular connectivity.

Users can select the type of emergency, add critical details, include GPS coordinates, and broadcast the SOS through the mesh.

Nearby Phones

Discover nearby devices and build the emergency mesh.

The app identifies nearby relay nodes using BLE / Wi-Fi P2P, enabling SOS messages to hop across devices until they reach a gateway.

Technology & Implementation

How ResQ-Mesh Works

Phase 1

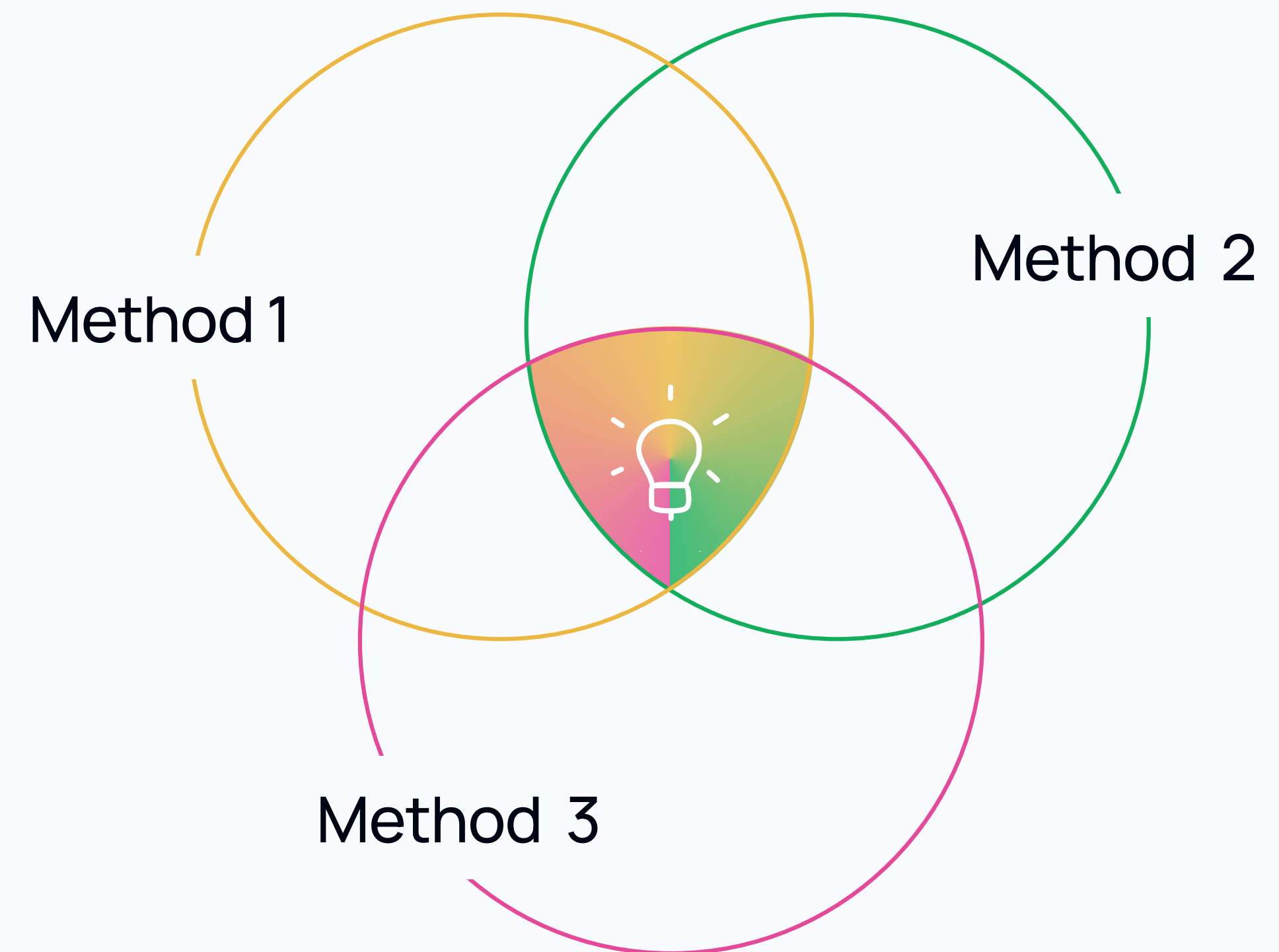
BLE + Wi-Fi Direct allow nearby smartphones to discover one another and relay encrypted SOS packets without cellular connectivity.

Phase 2

On-device AI analyzes distress signals such as cries, tapping, and structural impacts to estimate emergency priority locally.

Phase 3

Dynamic graph routing combines victim locations with mapped hazards to identify safer and more efficient rescue paths.





When the network goes down, we build a new one – device by device.

-Malini S



Thank You



ResQ-Mesh

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